Golden Games Implementation Brief

A step-by-step improvement plan for

`adbitrush.github.io/Elders_Ai`

Prepared for: handoff to Claude Code

Date: 26 August 2026

Site audited: live production build, 26 games, 6 languages

Scope: accessibility, architecture, retention, legal risk, SEO

1. How to use this document

This brief is written to be handed to Claude Code as a working specification. It is organised into **phases**, and each phase contains **tasks**. Every task has a goal, the work involved, and an acceptance test so the agent knows when it is done.

Section 9 contains ready-to-paste prompts, one per phase. Do not paste the whole document at once — work phase by phase, review the diff, commit, then move to the next. That keeps each change reviewable and lets you roll back a single phase if something breaks.

> **IMPORTANT LIMITATION OF THIS AUDIT**
>
> This brief was written from the *live rendered site*. The GitHub repository blocks automated access, so the source code was not read directly. Every claim about internal structure below is marked as either **observed** (visible in the live output) or **suspected** (needs verification). Phase 0 exists specifically so Claude Code verifies the suspected items against the real source before changing anything.

2. What the site is today

The honest summary: this is already well ahead of most of the senior-games market. The product instincts are right. What holds it back is architecture and polish, not concept.

Observed strengths

- **26 games**, all genuinely appropriate for the audience — memory, arithmetic, word, logic, attention, and two card games. That is a wider library than most competing sites.
- **Six languages** (Hebrew, English, Spanish, French, German, Greek) with Hebrew as the primary, RTL-first. Almost nobody serves Hebrew-speaking seniors. That is a real defensible niche.
- **Text-size control** in the header, plus sound toggle, stats, badges, and a profile. The accessibility instinct is already there.
- **No ads, no data collection** — stated in the footer. This is your single biggest competitive advantage over AARP and the 247-games network, and it should be marketed far harder than it currently is.
- **A difficulty selector** applied globally rather than per-game. Good idea, rarely done.
- Klondike explicitly built "בלי גרירה, רק לחיצות" — click-only, no dragging. That is exactly the right call for this audience and shows real understanding of the user.

Problems, ranked by how much they cost you

#	Problem	Status	Cost
1	Trademark exposure on two game names	Observed	Legal takedown risk
2	Single-page architecture — all 26 games in one document	Suspected	SEO, load time
3	Brand gold `#b7791f` fails WCAG contrast as text	Observed	Readability
4	26 JPEG card images, no evidence of lazy loading or WebP	Observed	Load time on old devices
5	No per-language URLs or hreflang	Observed	5 of 6 languages invisible to search
6	No daily-puzzle hook or streak mechanic	Observed	Retention
7	No printable output	Observed	Missing the highest-value niche feature
8	Not a complete installable PWA	Partial	Repeat visits
9	Developer-facing GitHub edit link in the footer	Observed	Confuses end users

3. Blocking issues — fix these first

3.1 Trademark exposure P0

> **ACT ON THIS BEFORE ANYTHING ELSE**
>
> **Tetris** is an actively enforced trademark. The Tetris Company is one of the most litigious rights-holders in games and routinely issues takedowns against browser clones — including non-commercial ones. Your game is currently named טטריס with an asset at `images/cards/tetris.jpg`.
>
> **Simon** is a Hasbro trademark. Your display name רצף צבעים ("colour sequence") is already safe, but the asset filename `simon.jpg` and any internal identifier `simon` should also be renamed.

To be clear about what is and is not at risk: the *mechanics* of falling-block puzzles and sequence-repetition games are not protectable. You may keep both games exactly as they play. What you must change is the **name**, the **filename**, and any **distinctive visual signature** associated with the branded original.

Current	Change to	Also rename
טטריס / Tetris	בלוקים נופלים / "Falling Blocks"	`tetris.jpg` → `blocks.jpg`; id `tetris` → `blocks`
asset `simon.jpg`	(display name already fine)	`simon.jpg` → `sequence.jpg`; id `simon` → `sequence`

Also review the falling-block piece colours. The classic cyan-I / yellow-O / purple-T colour scheme is a recognised trade dress. Use your own gold-and-earth palette instead — which also fits your brand better and improves contrast for older eyes.

This is general information about trademark risk, not legal advice. If the site ever earns revenue, have a lawyer review it.

3.2 Colour contrast failure P0

Your theme colour `#b7791f` measures a contrast ratio of **3.42:1** against white. WCAG requires **4.5:1** for normal body text. It passes only the 3:1 threshold that applies to large text (24px and above, or 19px bold).

For an audience where roughly one in six people over 70 reports vision impairment, this is not a technicality. The fix is a two-tone palette: keep the bright gold for large headings, borders, and decoration; use a darker gold for anything that is body-sized text.

Token	Value	Contrast on white	Use for
`--gold-bright`	#b7791f	3.42:1	Headings 24px+, borders, icons, decoration only
`--gold-text`	#8a5a00	5.93:1	Any text below 24px, links, labels
`--ink`	#1a1a1a	18.1:1	Body copy default
`--ink-muted`	#5c5c5c	7.0:1	Secondary text — never lighter than this

3.3 The PHP / GitHub Pages mismatch P0

ARCHITECTURE CONFLICT

GitHub Pages serves **static files only**. It does not execute PHP. Any `api/config.php`, admin token, or server-side searches viewer will be served to visitors as **plain text** — meaning any secret in that file is publicly readable by anyone who requests the URL.

Before Phase 1, confirm which is true:

- **(a)** The PHP lives on a separate host and only the static game front-end is on Pages. Fine — but make sure no PHP file was ever committed to the Pages repo.
- **(b)** PHP files are in the Pages repo. Then: remove them from the repo *and rotate the admin token immediately*, because it has been publicly served. Deleting the file is not enough — it remains in git history.

4. The design system

Everything below is calibrated to published accessibility research on older adults. Have Claude Code create `css/tokens.css` with exactly this, import it first, and refactor every hard-coded value in the codebase to reference these variables.

```
:root {
  /* --- Colour --- */
  --gold-bright: #b7791f;   /* 3.42:1 — large text 24px+ / decoration ONLY */
  --gold-text:   #8a5a00;   /* 5.93:1 — safe for all text sizes */
  --gold-deep:   #5c3a00;   /* headings on light backgrounds */
  --ink:         #1a1a1a;   /* 18.1:1 — body default */
  --ink-muted:   #5c5c5c;   /* 7.0:1  — never go lighter than this */
  --bg:          #fffdf7;
  --surface:     #ffffff;
  --border:      #d9c9a3;
  --success:     #2e7d32;
  --danger:      #b03030;
  --focus:       #0b5fff;   /* focus ring — must NOT be gold */

  /* --- Type: base is 20px, not 16px --- */
  --fs-base: 20px;
  --fs-lg:   24px;
  --fs-xl:   32px;
  --fs-2xl:  44px;
  --lh-body: 1.6;
  --lh-head: 1.25;
  --font: system-ui, "Segoe UI", "Noto Sans Hebrew", Arial, sans-serif;

  /* --- Touch targets & spacing --- */
  --tap-min: 60px;    /* minimum interactive element, both axes */
  --tap-gap: 16px;    /* minimum gap between two tappable things */
  --radius:  12px;
}

/* User-selectable text scale — the header 🔠 button sets data-scale on  */
html[data-scale="1"] { --fs-base: 20px; --fs-lg: 24px; --fs-xl: 32px; }
html[data-scale="2"] { --fs-base: 24px; --fs-lg: 30px; --fs-xl: 40px; }
html[data-scale="3"] { --fs-base: 30px; --fs-lg: 38px; --fs-xl: 50px; }

/* High-contrast mode toggle */
html[data-contrast="high"] {
  --bg: #000; --surface: #000; --ink: #fff;
  --ink-muted: #e0e0e0; --gold-text: #ffd24d; --border: #fff;
}

body { font-size: var(--fs-base); line-height: var(--lh-body);
       font-family: var(--font); color: var(--ink); background: var(--bg); }

button, a.btn, [role="button"] {
  min-width: var(--tap-min); min-height: var(--tap-min);
  font-size: var(--fs-lg); padding: 12px 20px;
}

:focus-visible { outline: 4px solid var(--focus); outline-offset: 3px; }

@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    transition-duration: 0.01ms !important;
    scroll-behavior: auto !important;
  }
}
```

Rules that go beyond the tokens

- **Keep everything important centred.** Peripheral vision declines substantially by age 70–80, so a meaningful share of your users effectively have tunnel vision. Do not put game state, scores, or controls in screen corners.
- **Never use colour alone to carry meaning.** Always pair it with a shape, icon, or text label. This matters doubly in the colour-word Stroop game.
- **Avoid red/green pairs** and do not rely on blue for contrast — blues appear faded to older eyes.
- **Sans-serif only.** No condensed faces, no all-caps.
- **"Back" and "Home" always visible.** חזרה לתפריט should be persistent and fixed, not something the user has to scroll to find.

5. Phase 0 — Audit before changing anything

Claude Code should complete this phase and report back *before* writing any code. The output is a written report, not a diff.

Tasks

1. **Inventory the repo.** Produce a file tree with byte sizes. Report total repo size, size of `index.html`, and the combined size of `images/cards/`.
2. **Confirm or refute the single-page hypothesis.** Are all 26 games in one HTML file, or are there separate pages with a shared shell? Report the actual routing mechanism.
3. **Measure the critical path.** How many bytes of HTML, CSS, JS, and images must download before the menu is usable on a cold load?
4. **Check for PHP or server-side files** anywhere in the repo, including git history (`git log --all --diff-filter=A --name-only`). Report anything found — see §3.3.
5. **Audit the i18n layer.** Where do the six languages live? One JS object, separate JSON files, or duplicated markup? Is the choice persisted? Does `<html lang>` and `dir` update on switch?
6. **List all hard-coded colours and font sizes** that will need to move to tokens.

7. **Check what state is stored** (scores, badges, profile, difficulty, text size) and where — localStorage, cookies, or memory only.
8. **Run an accessibility pass:** missing alt text, missing ARIA labels on icon-only buttons (🔠 🔊 📊 🏅), heading hierarchy, keyboard reachability of every game, focus visibility.

6. Phase 1 — Critical fixes

1.1 Rename trademarked games P0

Do: Apply the renames in §3.1 across display strings in all six language files, asset filenames, internal game IDs, and any saved-score keys. Add a one-time localStorage migration so existing users do not lose their high scores under the old key.

Accept when: `grep -ri "tetris\|simon" .` returns nothing outside of git history, and an existing user's saved scores survive the rename.

1.2 Apply the token system P0

Do: Create `css/tokens.css` from §4. Replace every hard-coded colour and font size. Raise base font from whatever it is to 20px. Enforce the 60px minimum tap target on every button and card.

Accept when: no hex colour literals remain outside `tokens.css`; every interactive element measures ≥60×60px in DevTools; automated contrast check reports zero failures.

1.3 Optimise images P1

Do: Convert all 26 card JPEGs to WebP with a JPEG fallback via `<picture>`. Resize to no more than 2× their displayed dimensions. Add `loading="lazy"` to every card below the fold and explicit `width` / `height` attributes to eliminate layout shift. Add real descriptive `alt` text in the active language — currently the alt appears to be just the game ID, which is useless to a screen reader.

Accept when: total image payload on first load is under 300KB; Cumulative Layout Shift is under 0.1.

1.4 Fix icon-button accessibility P0

Do: The header buttons (🔠 🔊 📊 🏅 ⭐) are emoji-only. Add `aria-label` in the active language to each, and `aria-pressed` to the toggles. Add a visible text label beneath each icon — for this audience, an unlabelled icon is a guess, not a control.

Accept when: every interactive element has an accessible name; a screen reader announces each header control meaningfully in Hebrew.

1.5 Remove the developer link P1

Do: The footer links to a GitHub edit page for `NOTES.md`. An 80-year-old who taps it lands on a login wall. Replace with a plain feedback route — a `mailto:` link is enough — or move it behind a keyboard shortcut only you know.

7. Phase 2 — Architecture

2.1 Split into per-game pages P1

This is the highest-leverage structural change. If Phase 0 confirms a single monolithic page, every game currently shares one URL — which means search engines see one page instead of 26, and every visitor downloads all 26 games to play one.

Do: Restructure to `/games/<id>/index.html`, one directory per game. Share the chrome via a small build step or by keeping the shell in a common JS module. Keep the menu at the root. Each game page gets its own `<title>`, meta description, and canonical URL.

Accept when: each game has a distinct shareable URL; the menu loads without downloading any game logic; browser Back returns to the menu correctly.

> **Watch for:** GitHub Pages serves from a project subpath (`/Elders_Ai/`), so all internal links must be root-relative to that base, not to `/`. Get this wrong and every link breaks. Consider a custom domain to remove the subpath entirely.

2.2 Per-language URLs and hreflang P1

Do: Move each language to its own path — `/he/`, `/en/`, `/es/`, `/fr/`, `/de/`, `/el/`. Emit reciprocal `hreflang` tags plus `x-default`. Set `<html lang>` and `dir="rtl|ltr"` correctly per language. Generate `sitemap.xml` covering every game × every language.

Why it matters: right now five of your six translations are invisible to search engines, because a language chosen by JavaScript on one URL is not a separate page as far as Google is concerned. Fixing this multiplies your indexable surface by roughly six.

2.3 Complete the PWA P1

Do: The Apple meta tags are present but the rest is missing. Add `manifest.webmanifest` (name, short_name, 192/512 icons, `display: standalone`, `start_url`, theme colour), and a service worker that caches the shell and all game assets for offline play.

Why it matters: a home-screen icon removes the entire browser-and-URL-bar layer, which is the single biggest source of confusion for this audience. Offline play also means the site works in a care home with bad wifi.

8. Phase 3 — Retention and reach

3.1 Daily puzzle P1

Do: Seed each game's random generator from the current date (`YYYYMMDD`) so every player worldwide gets the same puzzle each day. Add a "Puzzle of the day" section at the top of the menu. Implement a deterministic PRNG — `mulberry32` is four lines and sufficient.

```
function mulberry32(seed) {
  return function() {
    seed |= 0; seed = seed + 0x6D2B79F5 | 0;
    let t = Math.imul(seed ^ seed >>> 15, 1 | seed);
    t = t + Math.imul(t ^ t >>> 7, 61 | t) ^ t;
    return ((t ^ t >>> 14) >>> 0) / 4294967296;
  };
}
const today = Number(new Date().toISOString().slice(0,10).replace(/-/g,''));
const rng = mulberry32(today);
```

Why it matters: a daily reset is what converts a visitor into a habit. It is the mechanic behind essentially every successful puzzle property.

3.2 Streak counter P2

Do: Track consecutive days played in localStorage. Display prominently. Be gentle about breaking it — this audience finds aggressive streak-loss messaging genuinely upsetting. Use encouraging language, and consider a one-day grace period.

3.3 Printable puzzles P1

Do: Add a large-print print stylesheet and a "Print this puzzle" button on word search, sudoku, and the word games. Target 18pt minimum in print, pure black on white, no background images, answers on a second page.

Why it matters: this is the most under-served feature in the entire niche and it is nearly free to build. Printables get photocopied and passed around senior centres and care homes — it is the one feature that produces genuine offline word-of-mouth. It also serves the substantial part of your audience who prefer paper.

3.4 Market the no-ads position P2

Do: Your footer mentions no ads and no tracking. Move that to the top of the page as a prominent badge. Add a short privacy page stating plainly that nothing leaves the device.

Why it matters: the dominant competitors run pre-roll ads before play and crowded pages. Being visibly the calm, ad-free alternative is your entire differentiation — but only if visitors see it before they bounce.

9. Ready-to-paste prompts for Claude Code

Run these in order. After each one, review the diff and commit before continuing.

Prompt A — Audit

```
You are auditing the Golden Games codebase (a Hebrew-first brain-games site for
seniors, deployed to GitHub Pages). Do NOT change any code in this task.

Produce a written report covering:
1. Full file tree with byte sizes. Total repo size, index.html size, images/ size.
2. Is this a single-page app with all 26 games in one document, or separate pages?
   Describe the actual routing mechanism.
3. Cold-load critical path: bytes of HTML/CSS/JS/images before the menu is usable.
4. Any PHP or server-side files, including in git history:
   git log --all --diff-filter=A --name-only | sort -u
5. How i18n works across the 6 languages. Is the choice persisted? Do html lang
   and dir update on switch?
6. Every hard-coded colour and font-size, with file:line.
7. What is stored in localStorage/cookies (scores, badges, difficulty, text size).
8. Accessibility issues: missing alt, unlabelled icon buttons, heading order,
   keyboard traps, invisible focus.

Output as markdown. End with your top 5 recommendations ranked by impact/effort.
```

Prompt B — Phase 1

```
Implement Phase 1 of the improvement brief.

1. TRADEMARK RENAME (do this first, it is legal risk):
   - Rename the Tetris game to "בלוקים נופלים" (he) / "Falling Blocks" (en) and
     equivalents in es/fr/de/el. Rename internal id tetris -> blocks and asset
     images/cards/tetris.jpg -> blocks.jpg.
   - Rename asset images/cards/simon.jpg -> sequence.jpg and internal id
     simon -> sequence. Display name stays "רצף צבעים".
   - Change the falling-block piece colours away from the classic cyan-I /
     yellow-O / purple-T scheme to our gold/earth palette.
   - Add a one-time localStorage migration so existing high scores under the old
     keys are carried over, then the old keys removed.
   - Verify: grep -ri "tetris\|simon" . returns nothing in working tree.

2. DESIGN TOKENS:
   Create css/tokens.css with the :root block from the brief (I will paste it).
   Import it first. Replace every hard-coded colour and font-size with a token.
   Base font size becomes 20px. Enforce min 60x60px on every interactive element
   with min 16px gaps. Add a visible 4px non-gold focus ring. Add the
   prefers-reduced-motion block.

3. IMAGES:
   Convert all 26 card images to WebP with  JPEG fallback. Resize to max
   2x displayed size. Add loading="lazy" below the fold, explicit width/height on
   all. Replace the alt text (currently just the game id) with a real description
   in the active language.

4. ICON BUTTONS:
   The header buttons 🔠 🔊 📊 🏅 ⭐ are emoji-only. Add localised aria-label to
   each, aria-pressed on toggles, and a visible text label under each icon.

5. Remove the GitHub NOTES.md edit link from the footer; replace with mailto:.

Show me the diff for each numbered item separately before applying.
```

Prompt C — Phase 2

```
Implement Phase 2: architecture.

1. Split the monolith into per-game pages at /games//index.html. Share chrome
   via a common JS module or a small build step. Menu stays at root and must load
   WITHOUT downloading any game logic. Each page gets its own title, meta
   description, and canonical URL.
   CRITICAL: GitHub Pages serves this repo from the subpath /Elders_Ai/. All
   internal links must be relative to that base. Verify every link works on the
   deployed subpath, not just locally.

2. Move each language to its own path: /he/ /en/ /es/ /fr/ /de/ /el/. Emit
   reciprocal hreflang tags plus x-default. Set html lang and dir correctly per
   language. Generate sitemap.xml covering every game x every language.

3. Complete the PWA: add manifest.webmanifest (name, short_name, 192 and 512
   icons, display standalone, start_url, theme colour) and a service worker
   caching the shell and all game assets for offline play.

Test after each step that the site still works when served from a /Elders_Ai/
subpath. Commit each step separately.
```

Prompt D — Phase 3

```
Implement Phase 3: retention.

1. Add a deterministic daily puzzle. Use mulberry32 seeded with YYYYMMDD so all
   players get the same puzzle each day. Add a "Puzzle of the day" block at the
   top of the menu. Apply to word search, sudoku, number sequence, unscramble,
   and proverbs.

2. Add a streak counter in localStorage tracking consecutive days played. Display
   prominently. Use gentle, encouraging language when a streak breaks — this
   audience finds harsh streak-loss messaging upsetting. Include a one-day grace
   period.

3. Add a large-print print stylesheet plus a "Print this puzzle" button on word
   search, sudoku, and the word games. Print target: 18pt minimum, pure black on
   white, no background images, answer key on page 2.

4. Move the "no ads, no tracking" message from the footer to a prominent badge
   near the top of the page. Add a short privacy page stating plainly that no
   data leaves the device.
```

10. QA checklist

Run this after every phase. It is the definition of done.

Check	Pass condition
Contrast	Every text/background pair ≥4.5:1, or ≥3:1 if 24px+
Tap targets	Every interactive element ≥60×60px, ≥16px apart
Zoom	Usable at 200% browser zoom with no horizontal scroll
Text scale	All three scale levels work in every game without clipping
Keyboard	Every game fully playable by keyboard; focus always visible
Screen reader	Game state changes announced via `aria-live`; all controls named
RTL	Hebrew layout correct; no LTR bleed in mixed number/text strings
Cold load	Menu interactive in under 3s on simulated slow 4G
Old device	Works on iOS Safari 14 and Chrome 90 — test, do not assume
Offline	Previously visited games playable with network disabled
Reduced motion	No animation when `prefers-reduced-motion` is set
No dragging	Every game playable with taps alone — no drag required anywhere
Subpath	Every link works when served from `/Elders_Ai/`

11. What not to do

- **Do not copy another site's code, CSS, artwork, or branding.** Game *rules* for solitaire, sudoku, word search, memory, and hangman are public domain and free to implement — that part is completely legitimate, and it is what you have already done. Lifting a competitor's assets is a takedown and a de-indexing risk.
- **Do not add a login.** Account creation is the single biggest barrier for this audience. If you eventually want cross-device sync, use a short resume code the user writes down — no email, no password.
- **Do not add pre-roll or interstitial ads.** Being the site without them is your entire competitive position. If you must monetise, one static banner below the game.
- **Do not add time pressure to games that lack it.** Unlimited thinking time is a core requirement here, not a difficulty setting.
- **Do not introduce a heavy framework.** Vanilla JS with no build step keeps pages under 100KB and loading instantly on the eight-year-old iPads a large share of your users actually own.
- **Do not let Claude Code refactor all 26 games at once.** One phase, one review, one commit. A broken deploy on a static host means the site is simply down.

12. Suggested order of work

Order	Task	Effort	Payoff
1	Trademark renames	1 hour	Removes takedown risk
2	Verify/rotate any exposed admin token	30 min	Closes a security hole
3	Design tokens + contrast fix	Half a day	Readability for every user
4	Icon button labels	1 hour	Controls become usable
5	Image optimisation	2 hours	Load time on old devices
6	Split into per-game pages	1–2 days	SEO + load time; unlocks everything else
7	Per-language URLs + hreflang	Half a day	~6× indexable pages
8	Daily puzzle + streak	1 day	Retention
9	Printables	Half a day	Word-of-mouth in care homes
10	PWA + offline	Half a day	Repeat visits, removes browser confusion

A CLOSING NOTE ON STRATEGY

The competitive gap here is not features — you already have more games than most rivals. It is that the incumbents in this niche are large, ad-heavy, and English-only, while you are calm, ad-free, and Hebrew-first. Every decision above should be measured against whether it widens that gap or narrows it.

Realistically, ranking against AARP or the 247-games network on generic English keywords is very hard. Your winnable ground is Hebrew-language senior games, where you may have no serious competition at all, and the long-tail printable and themed-puzzle queries that nobody serves well. Fix the architecture so search engines can actually see the Hebrew and the other five languages, and that becomes reachable.